Eric Xu

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Education

University of California, Santa Cruz — Santa Cruz, CA
M.S. in Scientific Computing and Applied Mathematics — *2026 – 2027*
B.S. in Applied Mathematics, Highest Honors in the Major | GPA 3.77/4.00 — *2024 – 2026*

- **Graduate coursework:** Scientific Machine Learning, Applied Dynamical Systems, Machine Learning and AI in Genomics

Research Experience

TACS Lab, UC Santa Cruz — Santa Cruz, CA
Undergraduate Researcher (Advisor: Prof. Ashesh Chattopadhyay) — *Sep 2025 – Present*

- Built an AI ensemble pipeline (GenCast + FourCastNet3, rare-event resampling) for 3-week heat wave and cold outbreak forecasts; members reached the observed extreme in **6 of 7** cases vs **0 of 7** for NOAA CFSv2.
- Benchmarked against raw CFSv2 over **42 cases**: lowered closest-member error from 2.92 K to **2.28 K** (better in 38 of 42) and raised the 7-day critical success index from 0.30 to **0.56**.
- Ran each forecast in about 8 hours on one 8×H100 node (**~$156**), an estimated **5–23× cheaper** than direct sampling; scheduled jobs with SLURM on cloud H100s and PBS on NCAR Derecho.
- Building a diffusion model that downscales ERA5 (1°) to HRRR (3 km) temperature and wind, trained on **16,000** paired snapshots from 2015–2025.

Projects

Cross-Species Transfer Learning with DeepSEA | *PyTorch, scikit-learn* — Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on **~1M** mouse ENCODE DNA windows; beat training from scratch on **115 of 133** chromatin features (median AUROC 0.877 vs 0.865).
- Reached best validation loss in **5 epochs vs 12** from scratch; showed zero-shot transfer is near chance (AUROC 0.50), so fine-tuning is required.

4D-Var Data Assimilation with a Learned Inverse Operator | *PyTorch, L-BFGS* — Apr – May 2026

- Reimplemented a Google Research data-assimilation method in PyTorch with a team of 5 (senior capstone); trained a CNN inverse observation operator on **32,000** Lorenz-96 trajectories.
- Cut state-estimation RMSE by **38–75%** vs the standard 4D-Var baseline across noise levels, and reached **~3× lower** optimization cost on 2D Kolmogorov flow.

Council of Sumer | *Python, FastAPI, LLM agents, OpenTelemetry* — Oct 2026

- Built a hackathon system (team of 3) where **5 LLM agents** with conflicting goals run as separate processes, debate over AGNTCY SLIM, and vote, with runoff and human tie-break rules.
- Wrote the deliberation engine, tracing, and benchmark: across 16 seeded games the council kept **73%** of cities vs **56%** for a single LLM and starved **27% fewer** people.

Teaching Experience

Baskin School of Engineering, UC Santa Cruz — Santa Cruz, CA
Teaching Assistant: Differential Calculus for Economics; Ordinary Differential Equations — *Sep 2026 – Present*
Tutor and Reader: Linear Algebra, Discrete Mathematics, Calculus II — *Sep 2025 – Jun 2026*

- Rehired for 3 consecutive quarters as linear algebra tutor and reader; ran weekly tutoring and office hours and graded against instructor rubrics.

Skills

Languages: Python, C++, MATLAB, R, Bash, RISC-V assembly
ML and Scientific Computing: PyTorch, JAX, NumPy, Triton (GPU kernels)
HPC and Tools: SLURM, PBS, Linux, Git, SSH
Mathematics: Linear algebra, ODEs and PDEs, dynamical systems, numerical methods, probability and statistics